

A person in a dark blue shirt is holding a large cardboard box. Another person in a purple shirt is partially visible on the right side of the frame. The background is blurred, showing green foliage and a bright outdoor setting.

Sales and Distribution The Fulfillment Process

Learning Objectives

- ❑ Describe the organizational levels associated with the fulfillment process.
- ❑ List and explain the master data associated with the fulfillment process.
- ❑ Identify the key steps in the fulfillment process and the data, documents, and information associated with each step.
- ❑ Explain how and why fulfillment is integrated with other processes.

A Basic Fulfillment Process



Functionality

- ❑ Sales Support
- ❑ Sales
- ❑ Shipping and Transportation
- ❑ Billing
- ❑ Credit Management
- ❑ Foreign Trade

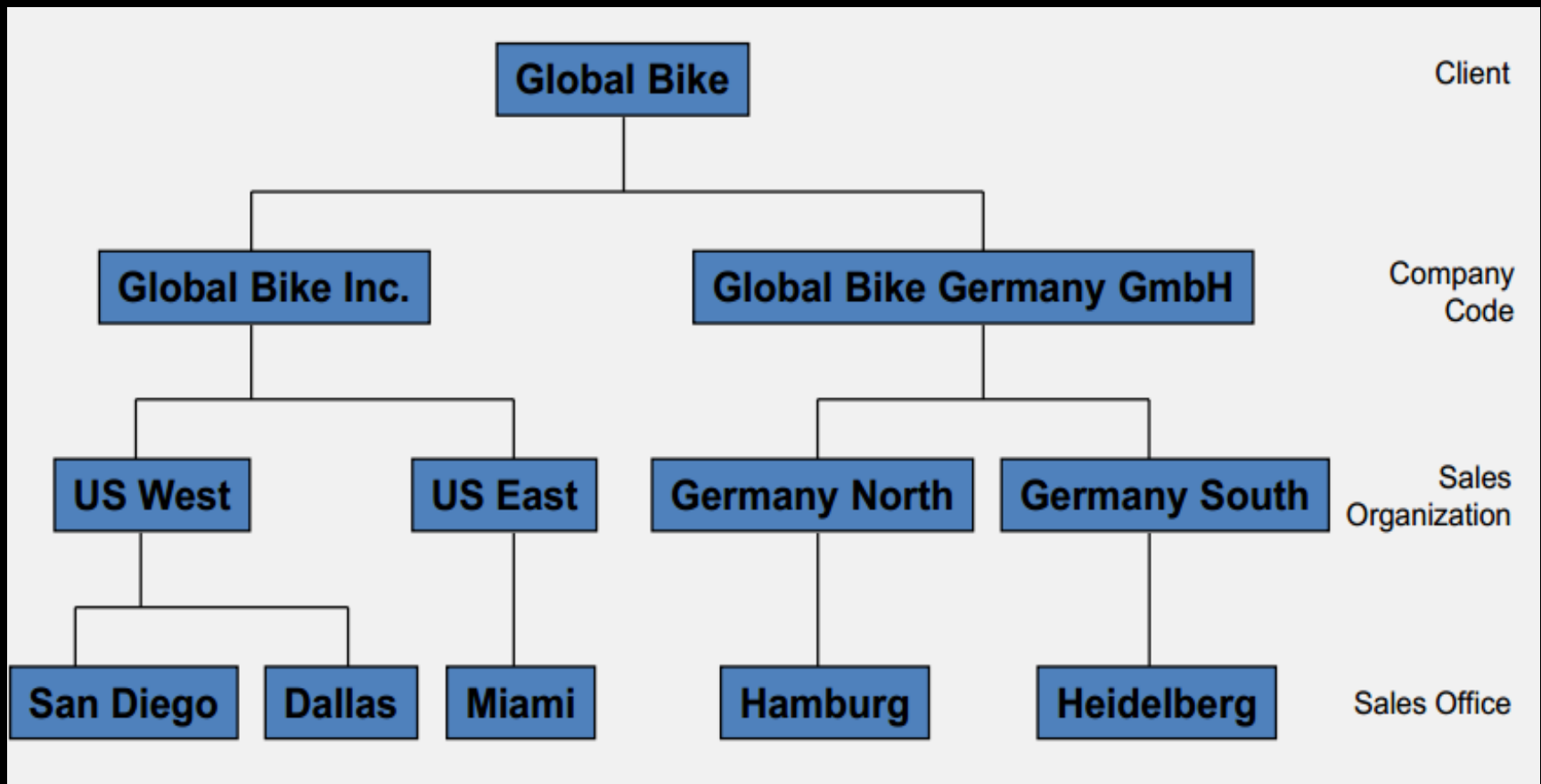
Organizational Data

- ❑ Client : An independent environment in the system
- ❑ Company Code : Smallest org unit for which you can maintain a legal set of books
- ❑ Credit Control Area: An organizational entity which grants and monitors a credit limit for customers.
- ❑ Sales Organization: An organizational unit responsible for the sale of certain products or services. The responsibility of a sales organization may include legal liability for products and customer claims
- ❑ Distribution Channel : The way in which products or services reach the customer. Typical examples of distribution channels are wholesale, retail, or direct sales

Sales Organization

- ❑ Division : A way of grouping materials, products, or services
- ❑ Sales Area: Combination of Sales Organization, Distribution Channel and Division. Determines conditions (i.e. pricing) for sales activities
- ❑ Plant (here: Delivering Plant): Plant from which the goods should be delivered to the customer
- ❑ Other SD organization units:
 - Shipping Point, Loading Point, Sales Office, Sales Group, Sales Person

GBI Sales Organizations



Master Data

- ❑ Customer master
- ❑ Material master
- ❑ Customer-material info record
- ❑ Conditions
- ❑ Output master data

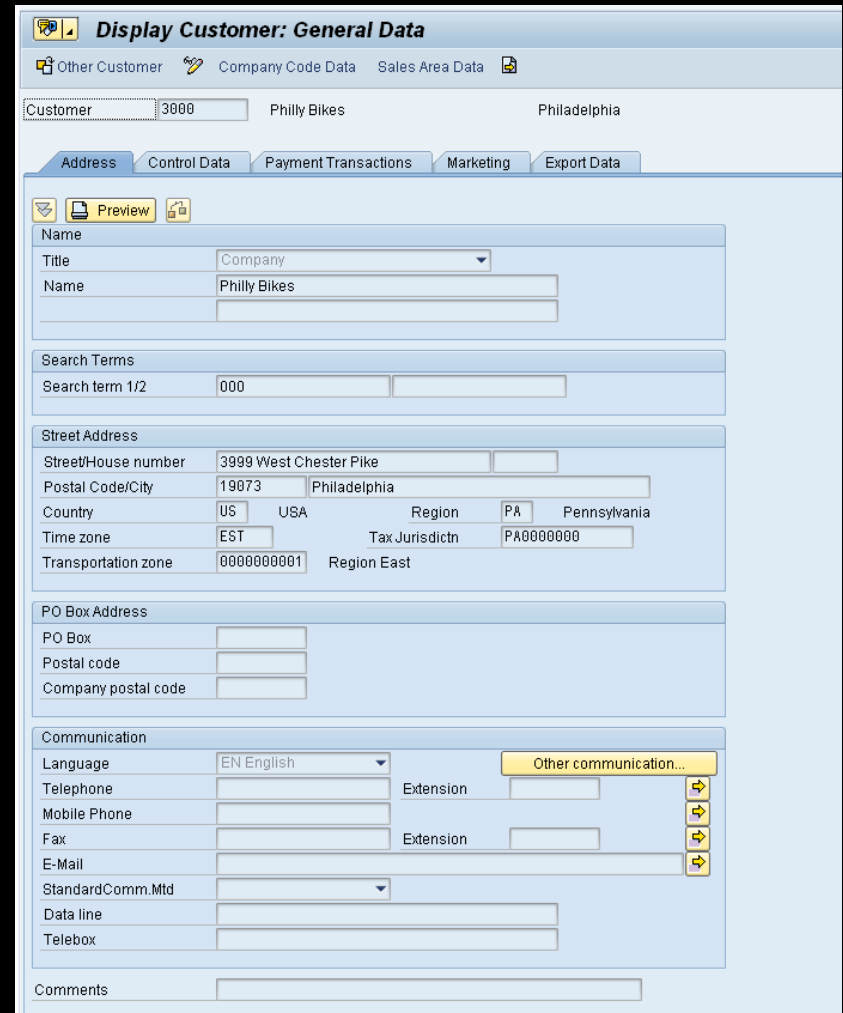
Customer Master Data

❑ Customer Master

- Contains all of the information necessary for processing orders, deliveries, invoices and customer payment
- Every customer **MUST** have a master record

❑ Created by Sales Area

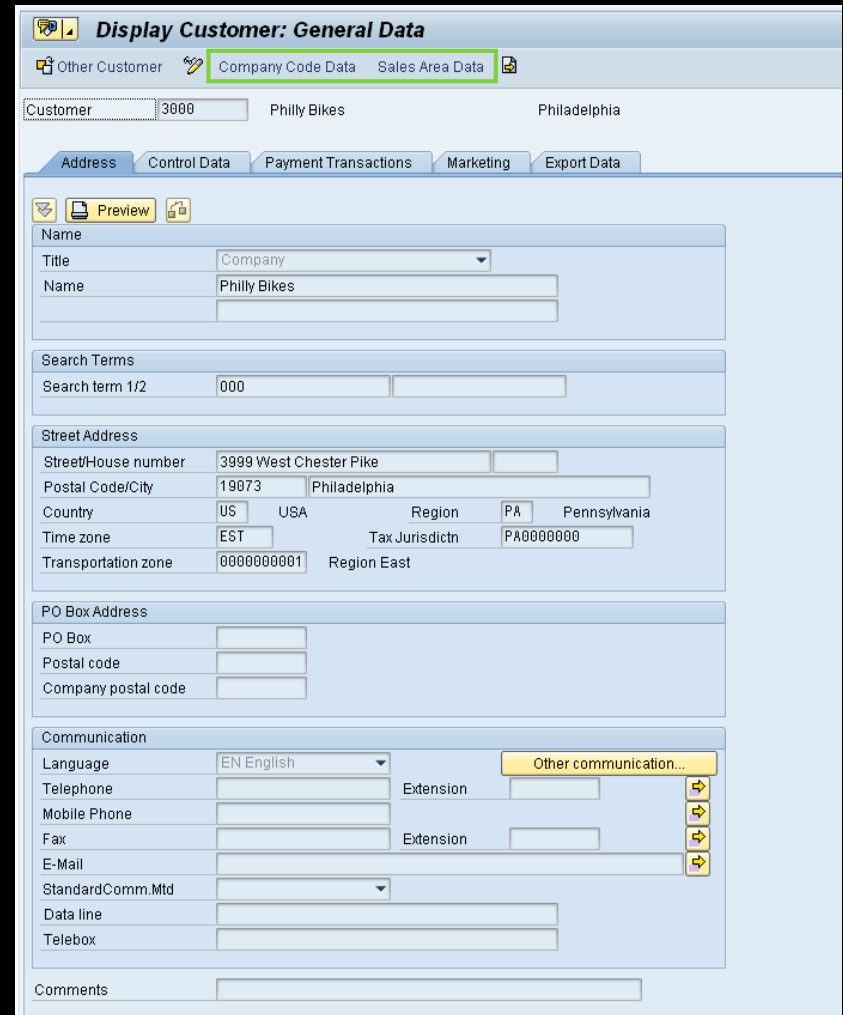
- Sales Organization
- Distribution Channel
- Division



The screenshot displays the 'Display Customer: General Data' form in SAP. At the top, there are tabs for 'Other Customer', 'Company Code Data', and 'Sales Area Data'. Below these, the customer details are shown: 'Customer' 3000, 'Philly Bikes', and 'Philadelphia'. The form is divided into several sections: 'Address' (with sub-tabs for 'Control Data', 'Payment Transactions', 'Marketing', and 'Export Data'), 'Search Terms', 'Street Address', 'PO Box Address', and 'Communication'. The 'Address' section includes fields for 'Name' (Title: Company, Name: Philly Bikes), 'Search term 1/2' (000), 'Street/House number' (3999 West Chester Pike), 'Postal Code/City' (19073 Philadelphia), 'Country' (US USA), 'Region' (PA Pennsylvania), 'Time zone' (EST), 'Tax Jurisdiction' (PA0000000), and 'Transportation zone' (0000000001 Region East). The 'PO Box Address' section has fields for 'PO Box', 'Postal code', and 'Company postal code'. The 'Communication' section includes fields for 'Language' (EN English), 'Telephone', 'Mobile Phone', 'Fax', 'E-Mail', 'Standard Comm. Mtd', 'Data line', and 'Telebox'. A 'Comments' field is at the bottom.

Customer Master Data (cont.)

- ❑ The customer master information is divided into 3 areas:
 - General Data
 - Company Code Data
 - Sales Area Data



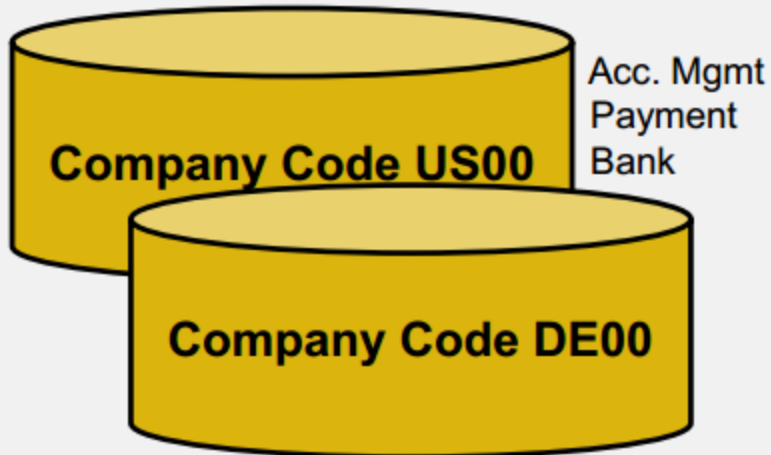
The screenshot displays the SAP 'Display Customer: General Data' form. At the top, there are tabs for 'Other Customer', 'Company Code Data' (which is highlighted with a green box), and 'Sales Area Data'. Below these, the customer number '3000' is entered, with 'Philly Bikes' and 'Philadelphia' as the company name and location respectively. The form is divided into several sections: 'Address' (with sub-tabs for 'Address', 'Control Data', 'Payment Transactions', 'Marketing', and 'Export Data'), 'Search Terms', 'Street Address', 'PO Box Address', and 'Communication'. The 'Street Address' section contains fields for 'Street/House number' (3999 West Chester Pike), 'Postal Code/City' (19073 Philadelphia), 'Country' (US), 'Region' (PA), 'Time zone' (EST), and 'Transportation zone' (0000000001). The 'PO Box Address' section has fields for 'PO Box', 'Postal code', and 'Company postal code'. The 'Communication' section includes fields for 'Language' (EN English), 'Telephone', 'Mobile Phone', 'Fax', 'E-Mail', 'Standard Comm. Mtd', 'Data line', and 'Telebox'. A 'Comments' field is at the bottom.

Customer Master Data (cont.)

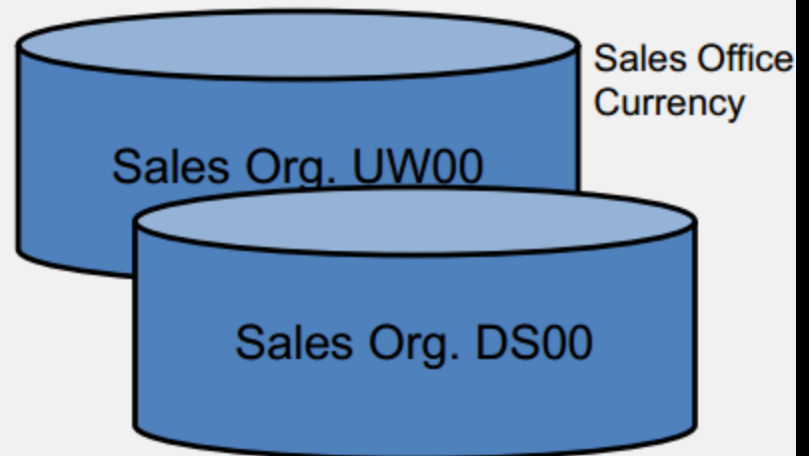
General Information relevant for the entire organization:



Company Code specific information:



Sales Area specific information:



Customer Master Data (cont.)

□ Partner Function

- Sold to party
- Ship to party
- Bill to party
- Payer

Material Master Data

❑ Material Master

- Contains all the information a company needs to manage about a material
- It is used by most components within the SAP system
 - ✓ Sales and Distribution
 - ✓ Materials Management
 - ✓ Production
 - ✓ Plant Maintenance
 - ✓ Accounting/Controlling
 - ✓ Quality Management
- Material master data is stored in functional segments called Views

Display Material DXTR1000 (Finished Product)

Additional Data Org. Levels

Basic data 1 Basic data 2 Sales: sales org. 1 Sales: sales org. 2 Sale...

Material DXTR1000 Deluxe Touring Bike (black)

General data

Base Unit of Measure	EA	each	Material Group	BIKES
Old material number			Ext. Matl Group	
Division	BI		Lab/Office	
Product allocation			Prod.hierarchy	
X-plant matl status			Valid from	
☐ Assign effect. vals			GenItemCatGroup	NORM Standard item

Material authorization group

Authorization Group

Dimensions/EANs

Gross Weight	8,510	Weight unit	6
Net Weight	8,510		
Volume	0.000	Volume unit	
Size/dimensions			
EAN/UPC		EAN Category	

Packaging material data

Matl Grp Pack.Matls

Basic Data Texts

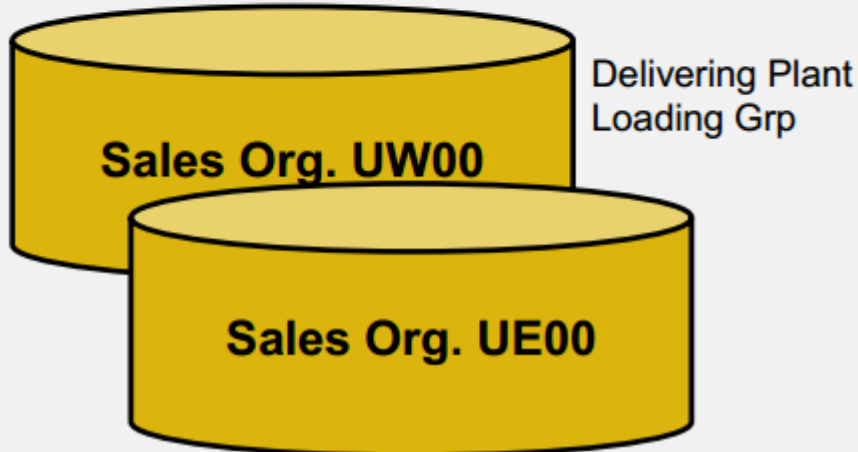
Languages Maintained 0 Basic Data Text Language:

Material Master Data (cont)

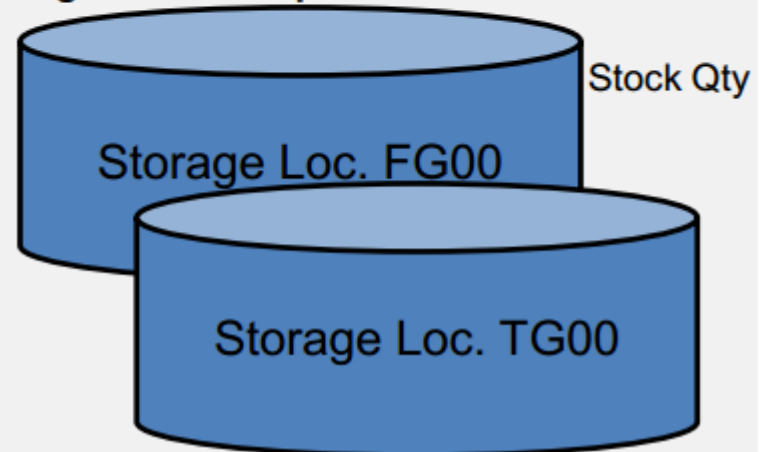
General Information relevant for the entire organization:



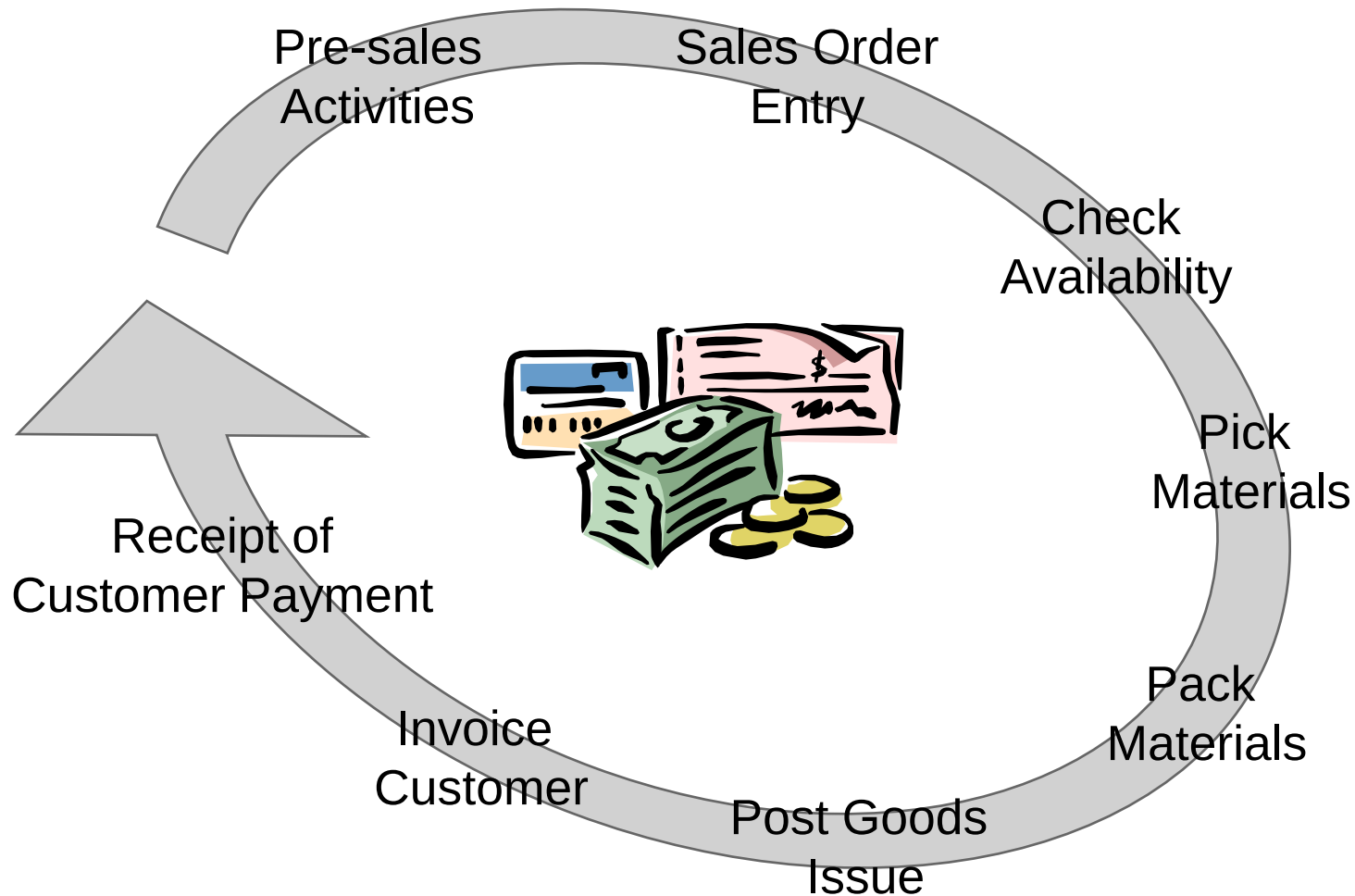
Sales specific information:



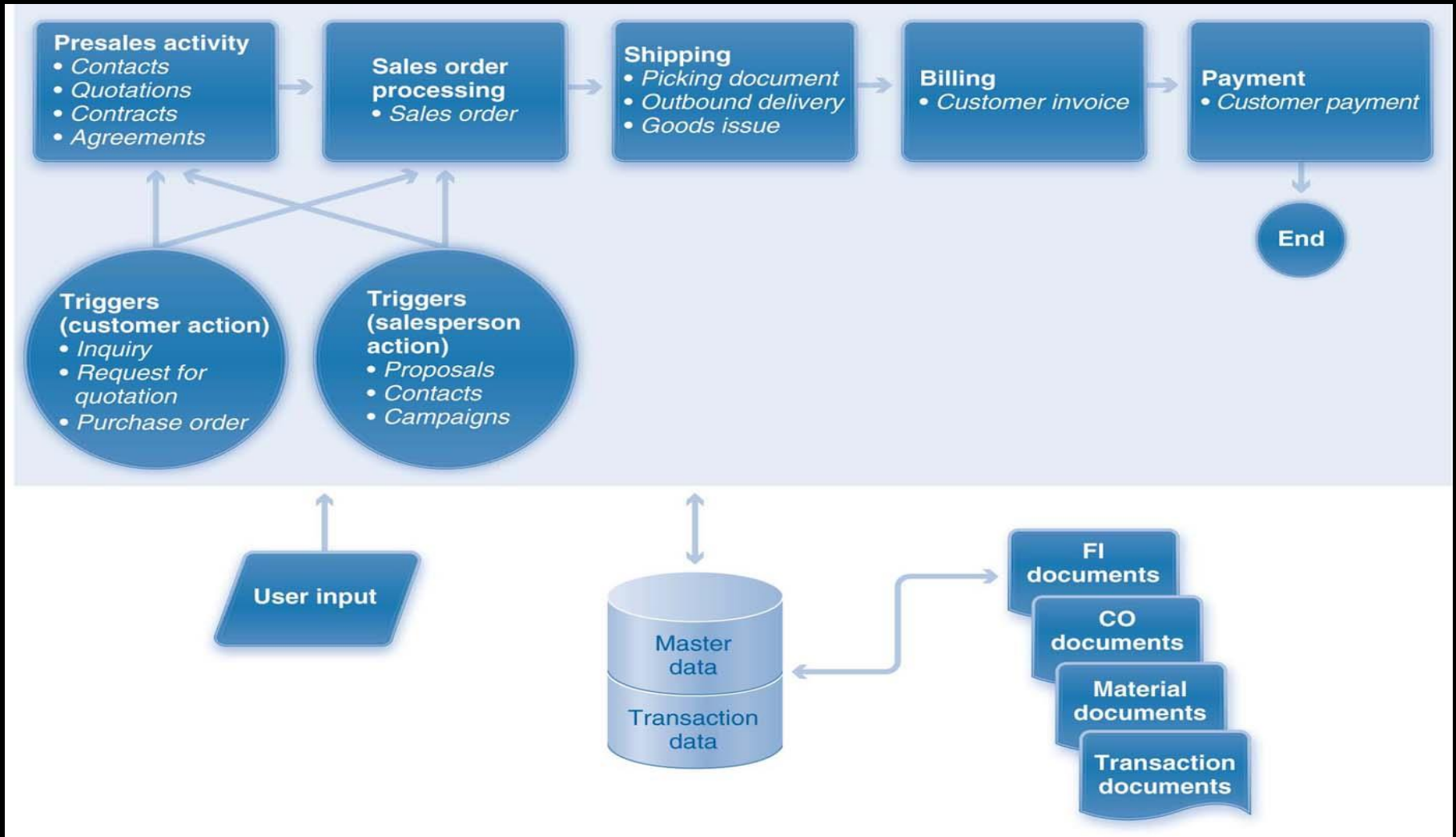
Storage Location specific information:



Fulfillment Process



Fulfillment Process Steps



Pre-Sales Activities

- ❑ The ultimate goal of all pre-sales activities is to equip the sales technician with all the information necessary to negotiate and complete the potential sale.
- ❑ Information needed:
 - Past sales activity
 - Past communication
 - Contact information
 - General Company info
 - Credit limits and usage
 - Current backorders
- ❑ 360° view of your customer

Sales activity

Type	Sales Call
From	10/04/2006 09:45
To	10/04/2006 10:22
Comment	Status HOT

Contact persons

Company	162	SAP, America // Newton Square PA
Contact person	Eva Romero	
Employee respons.	21	Grayson / 150 Forest Ave / US / 5

Follow-up actions

Create standard order [Start] [Link objects]

Info block: Key figures **View:** Statistical info **Customer:** 0000000162

Statistics measures						
	Value in T					
Document category	Min	Max	Totals	Average	Open	Number
Sales activities (CAS)	0	0	0	0		

Last SD documents			
Sales activities (CAS)	Date	Net value	Status
0100000003	03/10/06	0.00	Open
Order	Date	Net value	Status
0000000146	10/03/06	1,452.25 USD	Open
0000000145	09/25/06	1.09 USD	Open
0000000140	03/15/06	49,750.00 USD	Completed
0000000133	09/14/05	18,000.00 USD	Open
0000000132	09/14/05	13,500.00 USD	Being processed

Pre-Sales Activities

- ❑ Sales Support is a component of SD that assists in the sales, distribution, and marketing of a companies products and services to its customers. It contains the following functionality:
 - Creating and tracking customer contacts and communications (sales activity)
 - ✓ Phone call records
 - ✓ On-site meeting
 - ✓ Letters
 - ✓ Campaign communication
 - Implementing and tracking direct mailing, internet, and trade fair campaigns based on customer attributes
- ❑ Pre-sales documents need to be managed within the presales activities: **Inquiries and Quotations**. These documents help identify possible sales related activity and determine sales probability.

Inquiry

- ❑ An inquiry is a customer's request to a company for information or quotation in respect to their products or services without obligation to purchase.

- How much will it cost
- Material/Service availability
- May contain specific quantities and dates

- ❑ The inquiry is maintained in the system and a quotation is created to address questions for the potential customer.

Company The Bike Zone 2105 N Lewis Ave Portland OR 97227		Inquiry Number/Date 10000002 / 04/21/2010 Reference no./Date 000 / 04/21/2010 Delivery date Day 04/21/2010 Cust. no. 1301 Validity period 04/21/2010 bis 05/21/2010				
We deliver according to the following conditions: Terms of payment Payable immediately without deduction		Currency USD				
Terms of delivery FOB San Diego						
Weights (gross/net) - Volume - Mark Gross weight 57,170 G Net weight 57,170 G						
Please see our promotional offer enclosed. Delivery as long as stocks last.						
Item	Material	Qty	Description	Price	Price unit	Value
000010	DXTR1000	5 EA	Deluxe Touring Bike (black)	3,000.00	USD 1 EA	15,000.00
000020	PRTR1000	2 EA	Professional Touring Bike (black)	3,200.00	USD 1 EA	6,400.00
Final amount						21,400.00

Quotation

- The quotation presents the customer with a legally binding offer to deliver specific products or a selection of a certain amount of products in a specified timeframe at a pre-defined price.

Company		Quotation	
The Bike Zone		Number/Date	
2105 N Lewis Ave		20000000 / 04/21/2010	
Portland OR 97227		Reference no./Date	
		000 / 04/21/2010	
		Delivery date	
		Day 04/21/2010	
		Cust. no.	
		1301	
		Validity period	
		04/21/2010 bis 05/21/2010	

We deliver according to the following conditions: Currency USD

Terms of payment Payable immediately without deduction

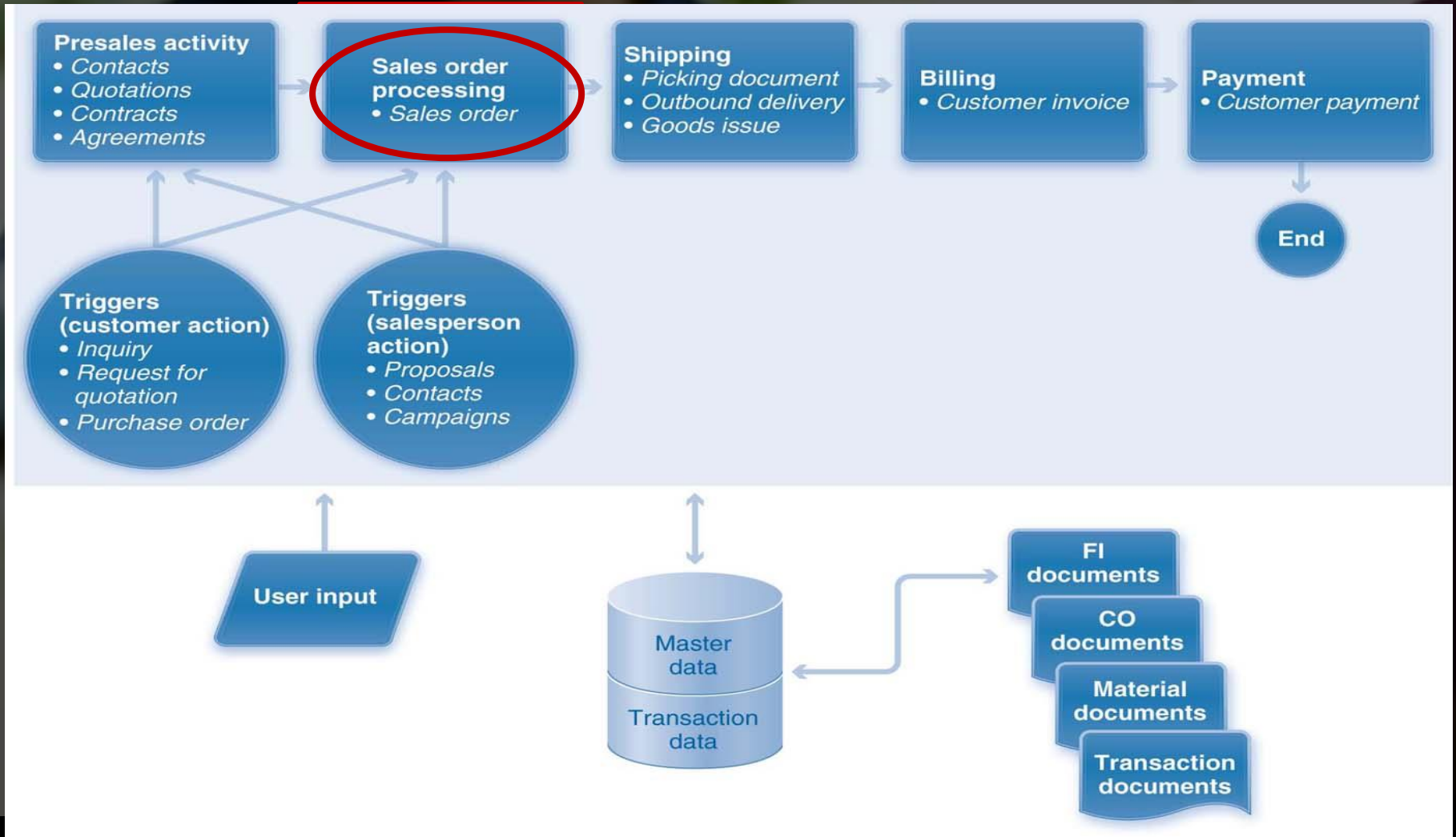
Terms of delivery FOB San Diego

Weights (gross/net) - Volume - Mark
 Gross weight 57,170 G Net weight 57,170 G

Please see our promotional offer enclosed. Delivery as long as stocks last.

Item	Material	Qty	Description	Price	Price unit	Value
000010	DXTR1000		Deluxe Touring Bike (black)			
		5 EA	3,000.00	USD	1 EA	15,000.00
	Material		50.00-	USD	1 EA	250.00-
	% Discount from N		5.000-	%		737.50-
	Net Value for Ite		2,802.50	USD	1 EA	14,012.50
000020	PRTR1000		Professional Touring Bike (black)			
		2 EA	3,200.00	USD	1 EA	6,400.00
	% Discount from N		5.000-	%		320.00-
	Net Value for Ite		3,040.00	USD	1 EA	6,080.00
Final amount						20,092.50

Fulfillment Process Steps



Sales Order

- ❑ Sales order processing can originate from a variety of documents and activities
 - Customer contacts us for order: phone, internet, email
 - Existing Contract
 - Quotations

- ❑ The electronic document that is created should contain the following basic information:
 - Customer Information
 - Material/service and quantity
 - Pricing (conditions)
 - Specific delivery dates and quantities
 - Shipping information
 - Billing Information

Sales Order (cont)

- ❑ The sales document is made up of three primary areas:

- Header

- ✓ Data relevant for the entire sales order: Ex: customer data, total cost of the order

- Line Item(s)

- ✓ Information about the specific product: Ex: material and quantity, cost of an individual line

- Schedule Line(s)

- ✓ Uniquely belongs to a Line Item, contains delivery quantities and dates for partial deliveries

Standard Order 15 Net value 20,092.50 USD

Sold-To Party 1301 The Bike Zone / 2105 N Lewis Ave / Portland OR 97227

Ship-To Party 1301 The Bike Zone / 2105 N Lewis Ave / Portland OR 97227

PO Number 000 PO date 04/21/2010

Tabs: Sales | Item overview | Item detail | Ordering party | Procurement | Shipping | Reason for rejection

Req. deliv. date D 05/21/2010 Deliver.Plant

☐ Complete div. Total Weight 57,170.6

Delivery block Volume 0.000

Billing block Pricing date 04/21/2010

Payment card Exp. date

Card Verif. Code

Payment terms 0001 Pay immediately w/o Incoterms FOB San Diego

Order reason

Item	Material	Order Quantity	Un	S	Description	Customer Material N
10	DXTR1000	5	EA	☐	Deluxe Touring Bike (black)	
20	PRTR1000	2	EA	☐	Professional Touring Bike (black)	

Sales Document Item 10 Item category TAN Standard Item

Material DXTR1000 Deluxe Touring Bike (black)

Tabs: Sales A | Sales B | Shipping | Billing Document | Conditions | Account assignment | Schedule lines | Pa

☐ Fixed date and qty Order Quantity 5 EA

Delivery time Delivered qty 5

Quantities/Dates						
P	Delivery Date	Order quantity	Rounded qty	Confirmed Qty	Sa... Delivery block	Delivered qty
D	04/21/2010	5	5	5	5 EA	5

Sales Order (cont)

- ❑ The sales order contains all of the information needed to process your customers request, the following information is determined for each sales order:
 - Delivering Schedule
 - Shipping point and route determination
 - Availability Check
 - Transfer of requirements to MRP
 - Pricing
 - Credit limit check

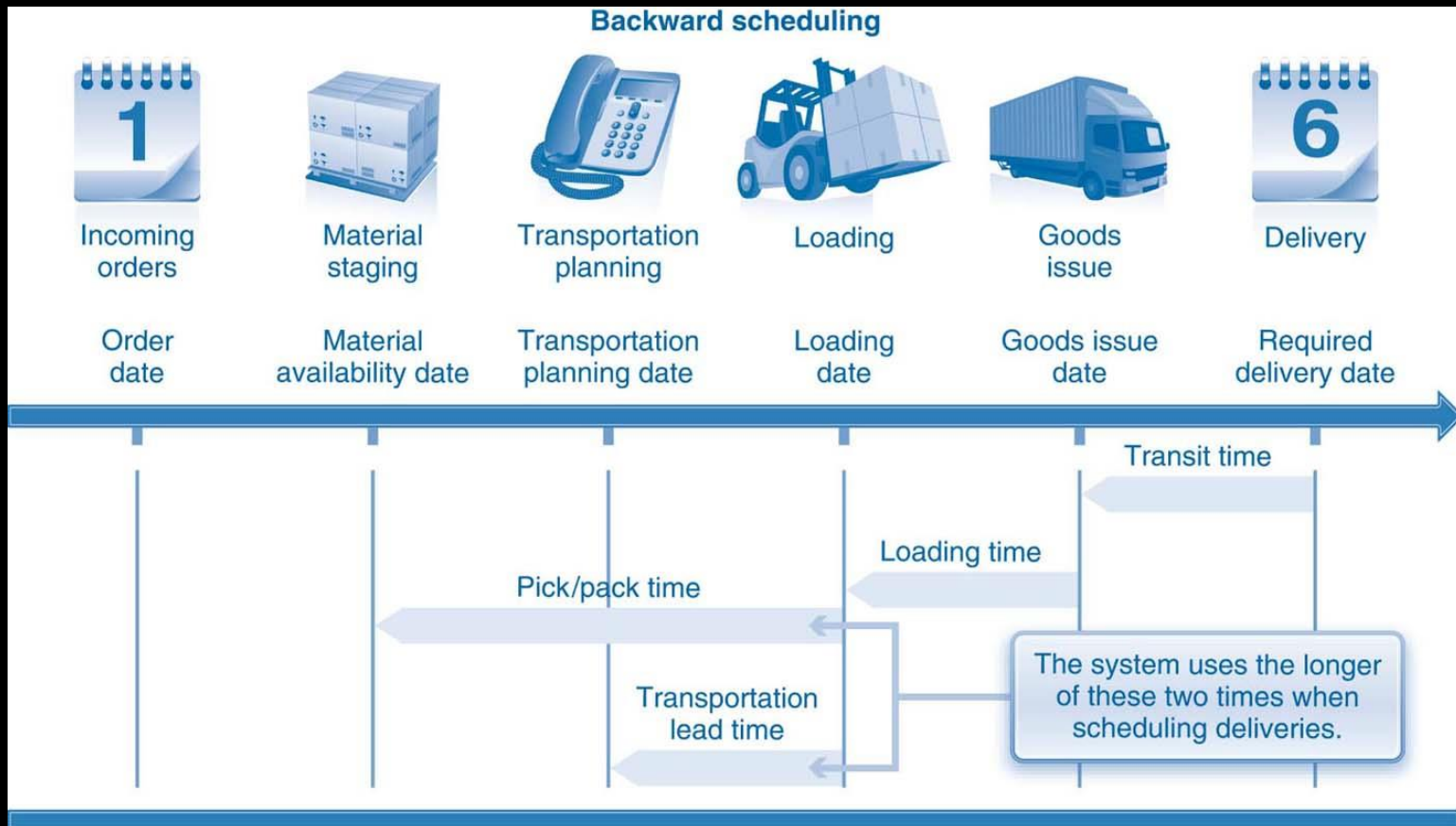
Relationship between quotation and SO



Delivery Scheduling

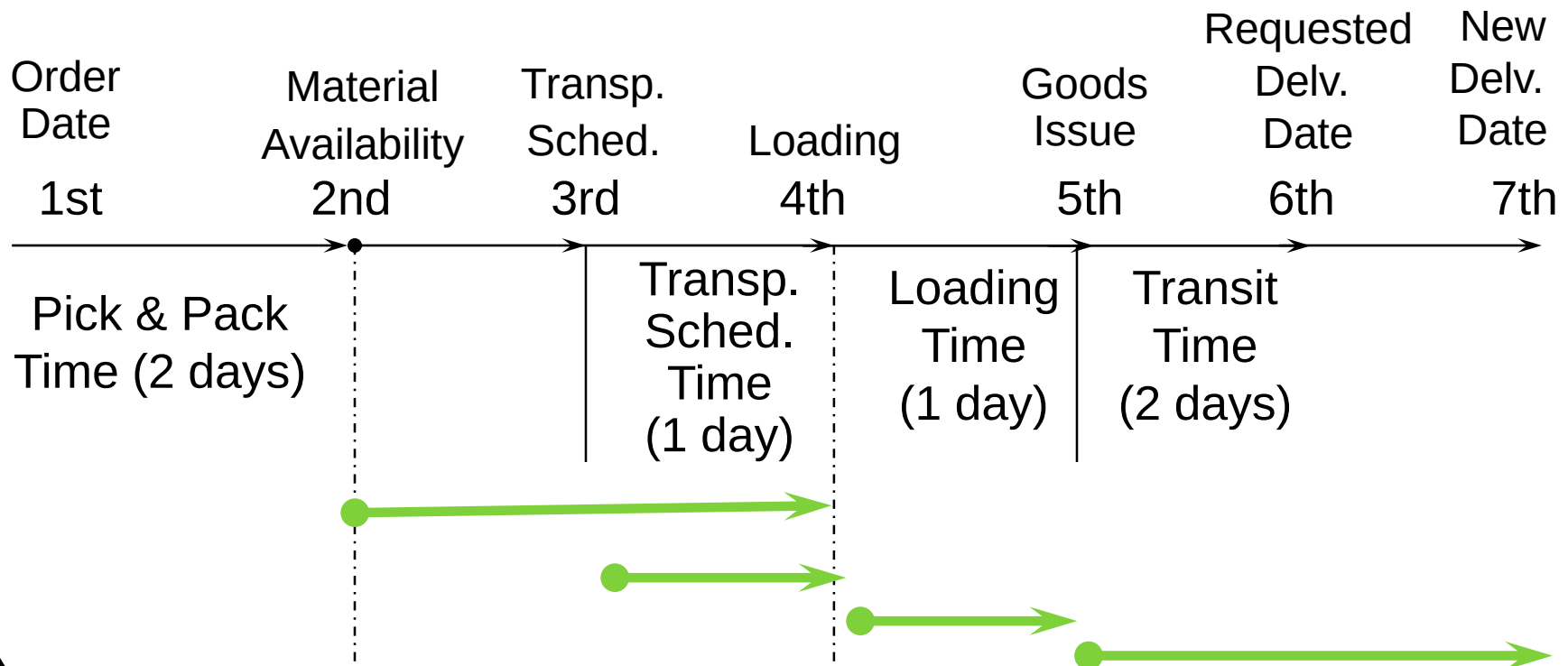
- ❑ When an order is created you must enter a requested delivery date for the order or each line item.
- ❑ The system will then determine a delivery timeline, this will be used when determining our material availability, or ATP (Availability to Promise) date.
- ❑ The system will determine this date using forward and backward scheduling rules you have defined.

Backward Scheduling



Forward Scheduling

Forward scheduling is also done if no product is available on the material availability date calculated by backward scheduling.



Availability Check

Availability Check

- Determines the material availability date
- Considers all inward and outward inventory movements
- ☐ Proposes 3 methods of delivery
- One-time delivery
- Complete delivery
- Delayed proposal
- ☐ Rules are created by YOU

The screenshot shows the 'Availability Overview' screen in SAP. At the top, there are icons for Stock, Totals records, and Scope of check. The main section displays the material 'DXTR1000' (Deluxe Touring Bike (black)) and the plant 'SD00'. The 'Avail. check' is set to '02' and the 'Check rule' is 'A'. The 'Base Unit' is 'EA'. Below this, the 'Totals display' section shows 'Receipts' as 75, 'Issues' as 8, and 'Confirmed issues' as 8. The 'ATP situation' table shows the following data:

Date	MRP e...	MRP element data	Rec./reqd qty
04/23/2010	Stock		75
04/23/2010	SLocSt	F600	75

Pricing

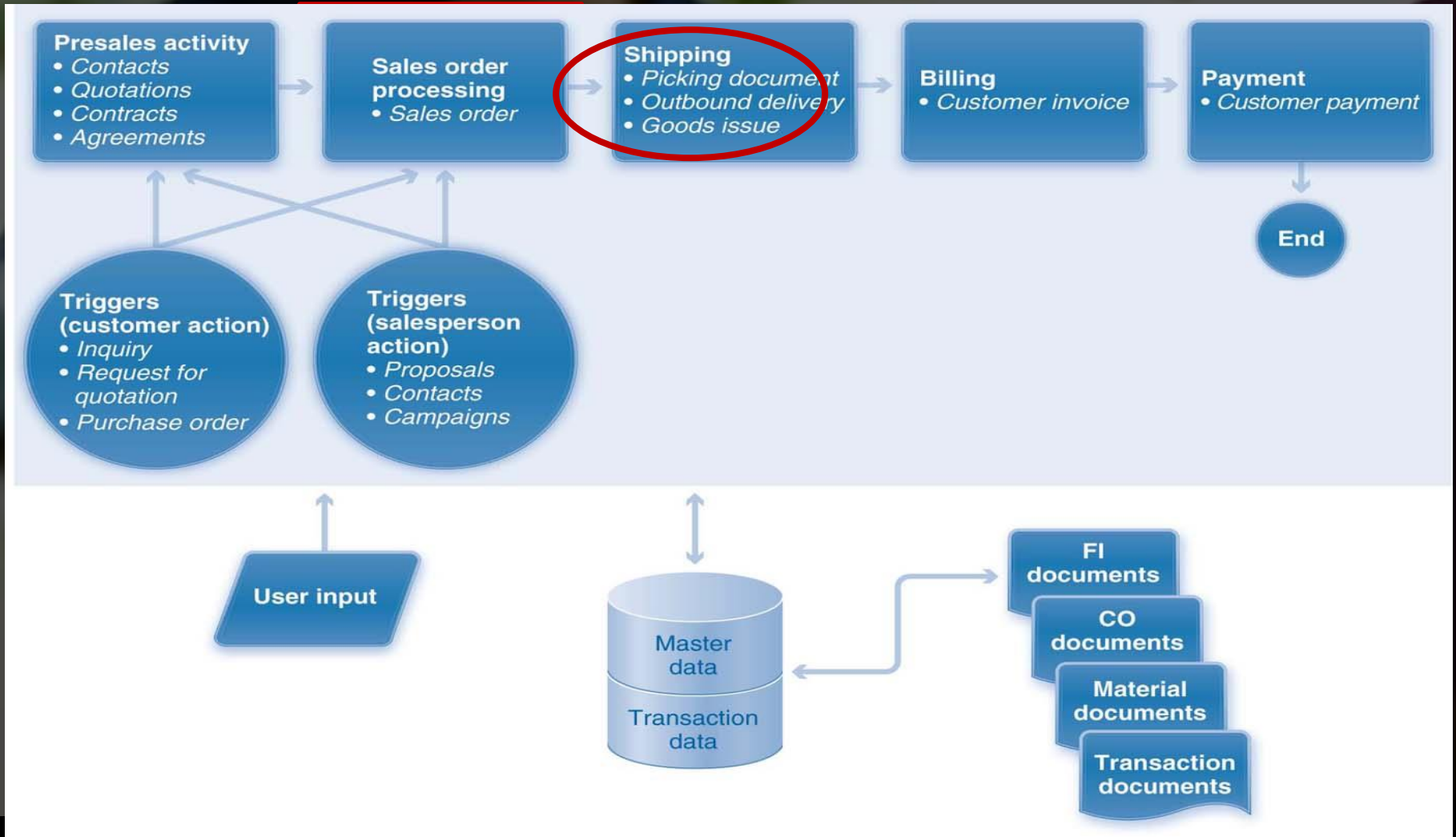
- The system displays pricing information for all sales documents on the pricing screens at both the header and the line item level.
 - Header pricing is valid for the whole order it is the cumulative of all line items within the order
 - Line item pricing is for each specific material.
- The system will automatically search for price, discounts, surcharges, calculate taxes and freight. You have the ability to manually manipulate the pricing at both the header and line item level within the sales order by entering a condition type and amount.
 - Taxes and freight can be set-up so we can't manually enter

Credit Check

- Allows your company to manage its credit exposure and risk for each customer by specifying credit limits.
- During the sales order process the system will alert the sales rep about the customers credit situation that arises, if necessary the system can be configured to block orders and deliveries.



Fulfillment Process Steps



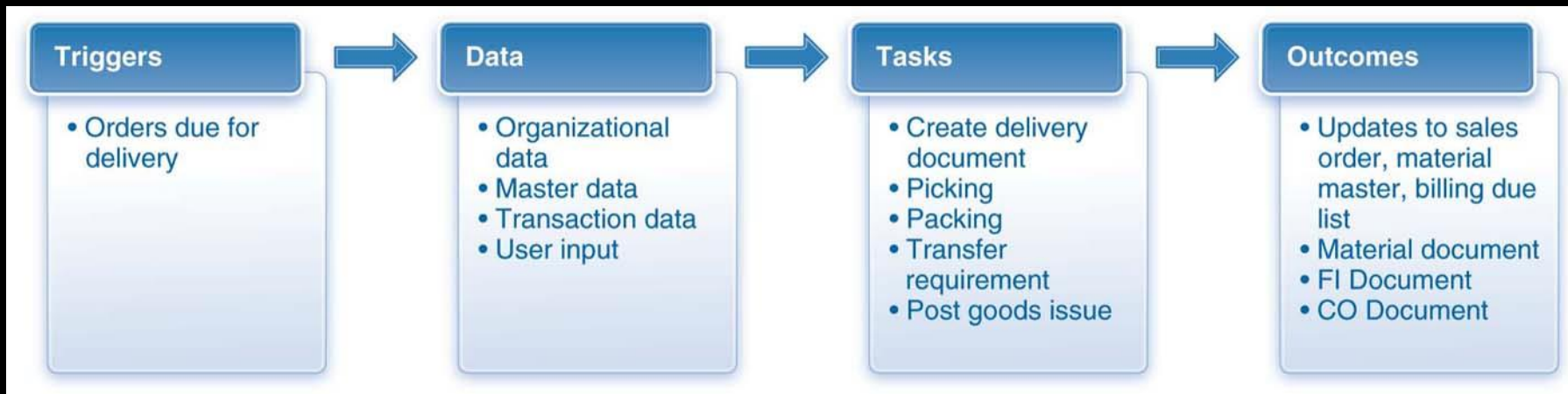
Shipping & Route Determination

- During the creation of the sales order the system must determine the shipping point from which the material will be shipped and the route the material will take to get from your warehouse to your customers location.
- A shipping point is determined for each line item within the order.
- The route determination is used to define the transit time of the material that we used in scheduling.

Shipping & Transportation

- ❑ The shipping process begins when you create the delivery document for the sales order. This document controls, supports, and monitors numerous sub-processes for shipping processing:
 - Picking
 - Packing
 - Post Goods Issue
- ❑ Integrated with the Material Management (MM) and Finance (FI) modules

Elements of the Shipping Step



Delivery Creation

- ❑ Checks order and materials to determine if a delivery is possible — delivery block (hold), completeness
- ❑ Confirms availability
- ❑ Confirms export/foreign trade requirements
- ❑ Determines total weight & volume

Delivery Creation

- ❑ Generates packing proposal
- ❑ Calculates weight and volume
- ❑ Checks scheduling
- ❑ Considers partial deliveries
- ❑ Updates route assignment
- ❑ Assigns picking location
- ❑ Updates sales order
- ❑ Determines batches
- ❑ Quality check (if needed)
- ❑ Updates sales order

Delivery Document

- ❑ The Delivery Document initiates the delivery process and is the control mechanism for this process
 - Picking
 - Packing
 - Loading
 - Posting Goods Issue
- ❑ Changes to delivery are allowable - products, quantities

Relationship SO and Delivery



Picking

- ❑ Quantities based on delivery note
- ❑ Assigned date when picking should begin
- ❑ Automated storage location assignment
- ❑ Supports serial number/lot number tracking and batch management
- ❑ Integrated with Warehouse Management (WM)

Delivery Qty vs Pick Qty



Loading and Packing

- ❑ Identifies which packaging is to be used for specified products (customer preference and UCC-128 considerations)
- ❑ Identifies and updates accounts associated with returnable packaging
- ❑ Tracks the packed product by container
- ❑ Insures weight/volume restrictions are enforced
- ❑ All packed items are assigned to the required means of transportation

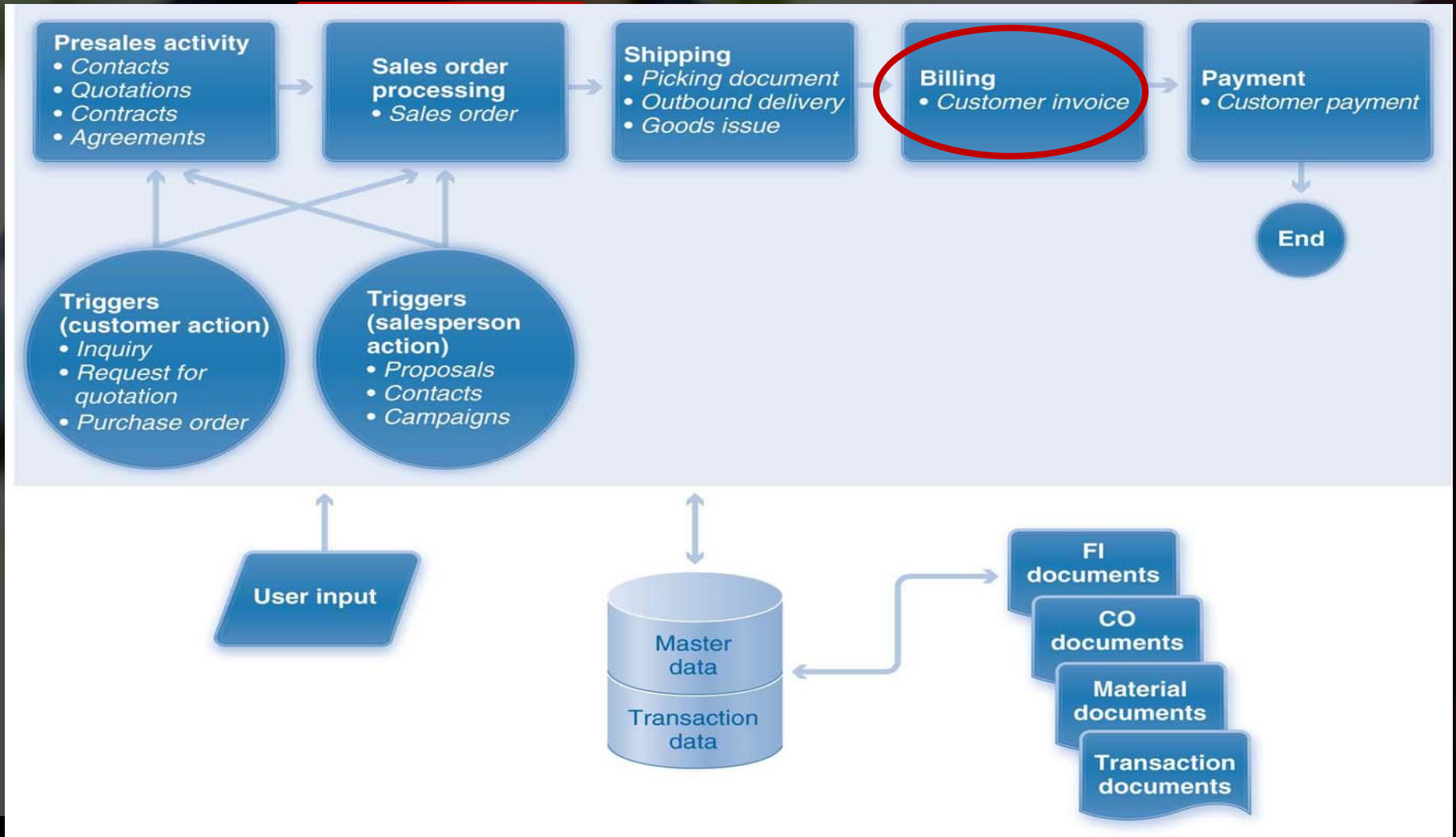
Goods issue

- ❑ Event that indicates the legal change in ownership of the products
- ❑ Reduces inventory and enters Cost of Goods Sold
- ❑ Automatically updates the General Ledger (G/L) accounts
- ❑ Ends the shipping process and updates the status of the shipping documents

Impact of Post Goods Issue

- ❑ Warehouse stock of the material is reduced by the delivery quantity
- ❑ Value changes are posted to the balance sheet account in inventory accounting
- ❑ Goods issue posting is automatically recorded in the document flow
- ❑ A work-list for the proof of delivery is generated
- ❑ After goods issue is posted for an outbound delivery, the scope for changing the delivery document becomes very limited. This prevents there being any discrepancies between the goods issue document and the outbound delivery.
- ❑ The billing due list is updated
- ❑ Sales documents are updated (status)

Fulfillment Process Steps



Billing

- ❑ The billing document is created by coping data from the sales order and/or delivery document.
 - Order-based billing
 - Delivery-based billing
- ❑ The billing process is used to generate the customer invoice.
- ❑ It will update the customer's credit status.

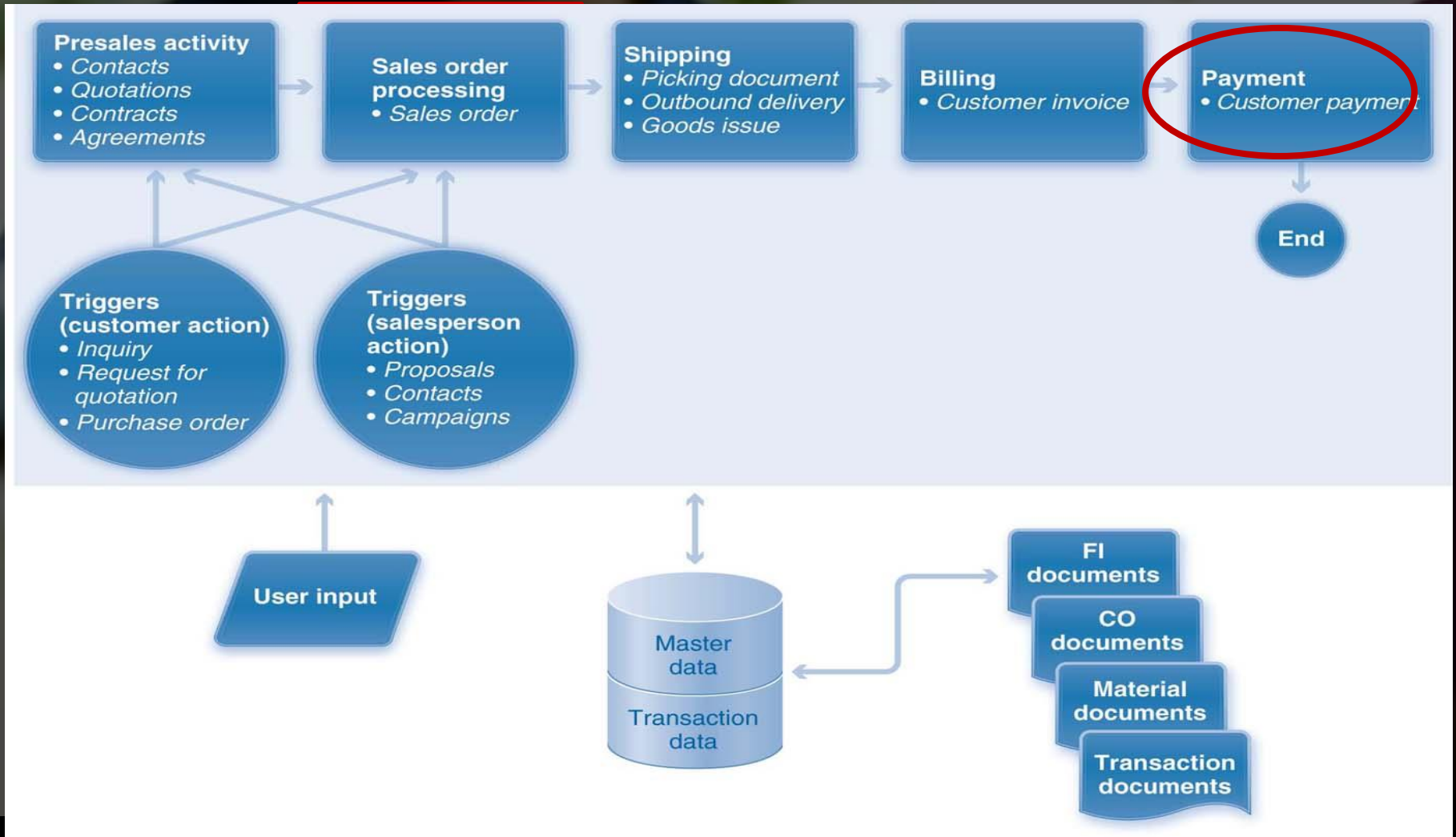
Billing Documents

- ❑ The billing document will automatically create a debit posting to your customer sub-ledger account and credit your revenue account.
- ❑ It is at this point that the sales process is passed over to Financial Accounting to await payment.

Relationship Delivery and Billing



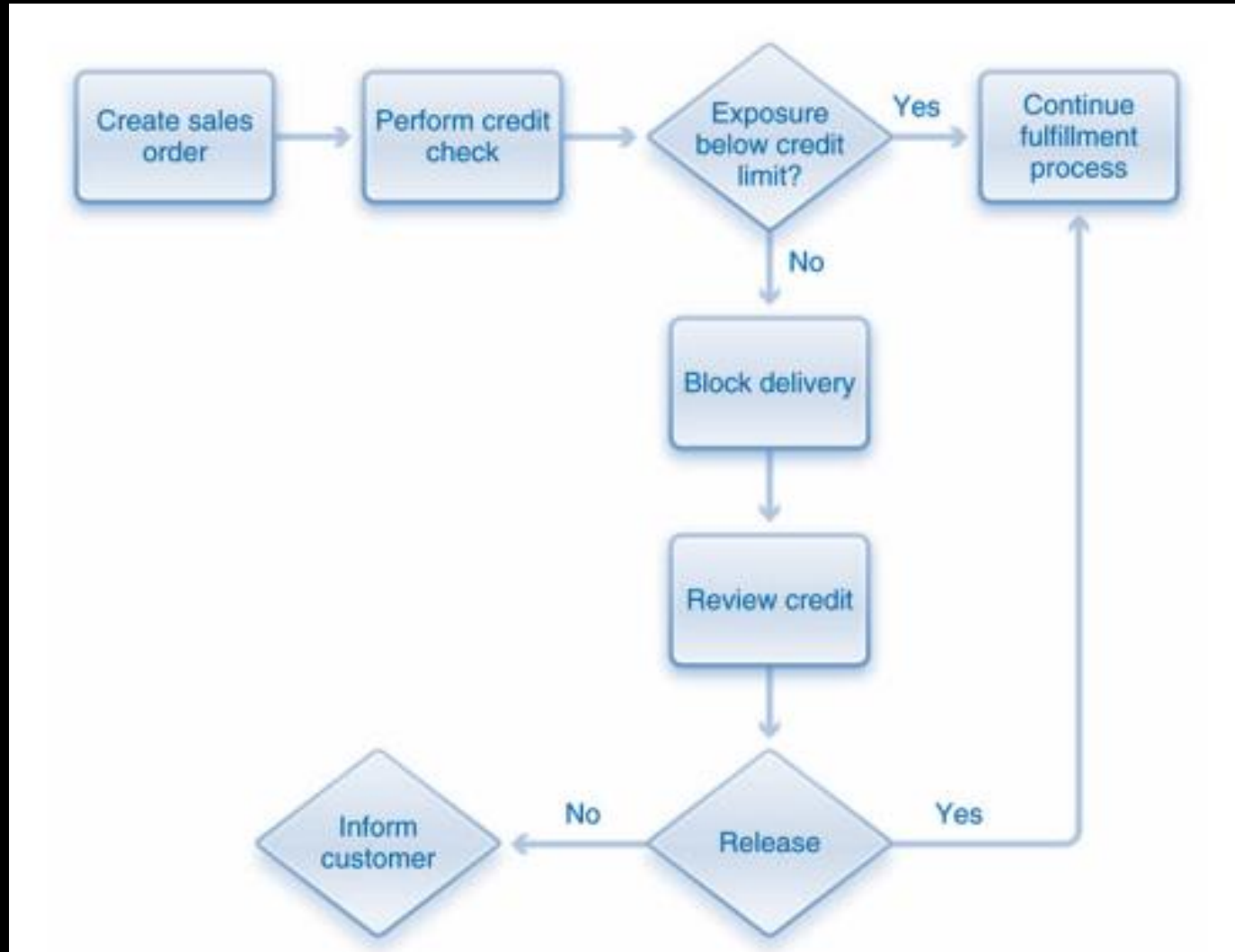
Fulfillment Process Steps



Payment

- ❑ Payment is the final step in the sales order process, this step is managed by the Financial Accounting department.
- ❑ Final payment includes:
 - Posting payments against invoices.
 - Reconciling differences between payment and invoice.
- ❑ Payment will create a posting clearing the liability in the A/R account and increases your bank account.

Credit Management Process



Integration with other process



Integration with other process

- ❑ Financial accounting
 - Because fulfillment involves both revenue and payments. Shipment, billing and payment have impact on General Ledger.
- ❑ Inventory Management, production, purchasing
 - Availability check
- ❑ Project System
 - Complex product like aircraft

References

- ❑ Textbook Chapter 5
- ❑ Kritchanchai, D., & MacCarthy, B. L. (1999). Responsiveness of the order fulfilment process. *International Journal of Operations & Production Management*, 19(8), 812-833.
- ❑ Croxton, K. L. (2003). The order fulfillment process. *International Journal of Logistics Management, The*, 14(1), 19-32.